TRANSPORTATION

GM to challenge China's LFP monopoly with upgraded battery factory

Tim De Chant · 9:04 AM PDT · July 14, 2025

GM said Monday it is working with joint venture partner LG Energy Solution to upgrade its Ultium battery factory to make lithium-iron-phosphate cells for the automaker's low-cost EVs.

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Spring Hill to produce LFP cells later this year, with commercial production expected by late 2027.

The automotive industry has been gravitating to LFP batteries in recent years for their low cost and appealing safety profile. The raw materials that make up the chemistry are cheaper and more widely available than other chemistries like nickel-manganese-cobalt (NMC), and they're also less likely to catch fire.

GM has been pursuing a three-pronged approach to battery sourcing for its EV range, which currently spans 12 models.

At the high end, it will continue using NMC in models that require longer range. The longest-legged Chevy Silverado EV, for example, uses a massive 205 kilowatt-hour NMC pack to drive 492 miles on a single charge.

In the middle, GM has developed a new chemistry it calls lithium-manganese-rich, or LMR. The new chemistry slashes the amount of nickel and cobalt in a cell and replaces it with cheaper, domestically available manganese. The result, GM said, will be a pack that delivers more range than LFP at a cost that's comparable to LFP today. LMR cells should hit the market in 2028.

For entry-level vehicles, GM is planning to switch to LFP, Andy Oury told TechCrunch in May. Given how consistently battery pack costs have come down over time, it's likely that LFP costs will drop below those for LMR in the coming years. The Spring Hill upgrades should start producing LFP cells for commercial sale in 2027.

Topics: Climate electric vehicles GM lfp LG lg energy solution lithium ion batteries Transportation Ultium

Tim De Chant

Senior Reporter, Climate

Tim De Chant is a senior climate reporter at TechCrunch.
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GM's new 'manganese-rich' battery promises cheaper EVs in 2028

Tim De Chant — 7:00 AM PDT · May 13, 2025

General Motors revealed Tuesday a new battery chemistry called lithium-manganese-rich (LMR), which it says should slash costs while delivering range that's just shy of the most advanced batteries

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less). GM is planning a version with cheaper
lithium-iron-phosphate (LFP) cells, which would

cracked the problem. The automaker experimented with a range of materials and manufacturing processes to arrive at the current formulation.

LMR ba
instead
the latte Kelly said switching to prismatic cells, which have a rigid shell, will help the company build a battery pack with more than 50% fewer parts.

"It's a huge, huge cost savings we'll get," he said.

GM has big plans for LMR, with the chemistry potentially spreading throughout the EV lineup. Andy Oury, business planning manager at GM, said that LMR could "take up a huge chunk in the middle" of the market, pushing LFP to entry-level vehicles and pricey NMC to applications that need long range and high energy density.

The new cells will be made by Ultium Cells, GM's joint venture with LG Energy Solution. Through Ultium, the two companies have invested billions of dollars in battery manufacturing in the United States.

Both have been pursuing LMR for years. GM has more than 50 patents on LMR, though LG has also been working on the technology itself. Kelty acknowledged it's possible that LG could make its own version of LMR cells that don't infringe on GM's patents, making the chemistry more widely available. "It'll be interesting to see how this all plays out," Kelty said.

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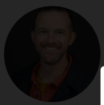
to modify its existing manufacturing plants to accommodate the new chemistry and then scale up production. Scaling, in particular, tripped up the first Ultium cells.

Kelty is confident that GM can hit the 2028 target.

"It meets all our performance metrics, we have a partner that's going to manufacture it, and we've got a manufacturing location," he said. "The other thing is, the supply chain is much more local than high-nickel or LFP, so we're really incentivized to do this. There's a lot of things coming together here that really make us want to go quickly."

Topics: battery manufacturing Climate critical minerals electric vehicles GM lg energy solution lithium ion batteries Transportation



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Ford adds battery capacity, LFP chemistry to scale global EV business

Jaclyn Trop 6:40 AM PDT · July 21, 2022

Ford said Thursday it has shored up its battery supply chain as part of its global strategy to sell more than 2 million EVs annually by 2026.

The automaker, which is investing $50 billion to scale its battery-electric portfolio through 2026, said it plans to boost battery ... bartop its supply ... batterie... less ex... among ... burgeo...

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Meeting its EV production goals could require about [8,000 job cuts, according to unnamed sources cited by Bloomberg.](#)

Executives did not address layoffs directly during a briefing with analysts and media Thursday but said that speed and agility is crucial for Ford's battery-electric Model e division. Cutting out bureaucracy means they can close deals in days, rather than months, said Lisa Drake, vice president of EV industrialization at Ford Model e.

"Believe it or not to move fast in this space, smaller is better," she said. "A smaller team can move

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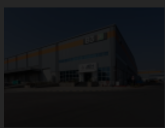

Jeh Aerospace nets $11M to scale the commercial aircraft supply chain in India

Jagmeet Singh - 2 days ago

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Foxconn sells former GM factory to mystery buyer after failing to make EVs

Sean O'Kane - 3 days ago

faster than a larger team."

Lithium iron phosphate batteries

Ford said it will begin incorporating LFP batteries into its portfolio while continuing to use the nickel cobalt manganese (NCM) chemistry currently powering its EVs. As the cheaper alternative, LFP presents a potential path for the mass EV market, according to analysts.

CATL, the world's largest battery supplier, will provide LFP battery packs for Ford Mustang Mach-E SUVs in North America starting next year, followed by F-150 Lightning pickup trucks in early 2024.

LFP batteries provide cost advantages over NCM but are less powerful due to their lower energy density. The chemistry may seem a puzzling choice to underpin two of Ford's most muscular models, but "these standard-range batteries offer customers many years of operation with minimal loss of range after many charge cycles," Drake said. "That benefits owners who need to charge often, like our commercial customers."

Using LFP will help reduce Ford's reliance on scarce critical minerals such as nickel and cobalt, while trimming 10 to 15% from its battery materials costs, according to Drake.

The company declined to say how many of its EVs will use LFP chemistry but said it has signed non-binding MOUs with lithium suppliers Liontown Resources and Rio Tinto.

Batt

Ford also said it has upped its contracts with NCM battery suppliers in order to meet its late-2023 production target. LG Energy will double its NCM battery capacity to supply Ford's Mach-E and E-Transit van.

SK On has increased production at its Georgia facility to supply Ford with more NCM cells. It will also supply additional battery packs from its factory in Hungary to power the F-150 Lightning and E-Transit through 2023.

That's in addition to the deal Ford and SK On closed last week to create an $11.4 billion joint venture to build and operate the Blue Oval City complex in Stanton, Tennessee, and two EV battery plants in Glendale, Kentucky.

Ford also announced Thursday a non-binding agreement with CATL to explore partnering on LFP batteries in Chinese and European markets.

Raw Materials

As Ford scales its EV production, it must reinvent its supply chain by sourcing raw battery materials directly

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Jaclyn Trop
Reporter, Transportation

Jaclyn Trop covers EVs and automotive technology at
TechCr

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